

KATHLEEN (KATY) ABBOTT

Woods Hole Oceanographic Institution
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EDUCATION

MIT-WHOI Joint Program in Oceanography Cambridge, MA
Ph.D. in Physical Oceanography 2026

Thesis title: Vertical transport and transformation of biogeochemical properties in the upper ocean

University of California, Berkeley Berkeley, CA
B.A. in Geophysics, Applied Mathematics with high honors 2017

PROFESSIONAL APPOINTMENTS

Foster and Coco Stanback Postdoctoral Fellow Pasadena, CA
California Institute of Technology Starting Oct. 2026

Endless Frontiers Fellowship Postdoctoral Fellow Woods Hole, MA
Woods Hole Oceanographic Institution 2026 - present

GIS Technician Big Bend, TX
Big Bend National Park 2019 - 2020

Post-bac Helen Fellow New York, NY
American Museum of Natural History 2018 - 2019

FELLOWSHIPS & GRANTS

Caltech Foster and Coco Stanback Postdoctoral Fellowship 2026-2028
in Environmental Science and Engineering

WHOI Endless Frontiers Postdoctoral Fellowship 3-month fellowship award 2026

Ocean Ventures Fund to measure bacterial respiration in the Bay of Bengal **\$9710**, 2024-2025

National Science Foundation Graduate Research Fellowship 2020-2025

MIT Ida M. Green Fellowship 2020-2021

American Meteorological Society Graduate Student Fellowship 2020-2021

PUBLICATIONS

Pre-prints

1. Spezzano, R., Matantseva, O., **Abbott K.**, Hauss, H., Wittmers, F., Riebesell, U., Achterberg, E., Browning, T.J., Mahadevan A., Worden, A.Z. (in prep) *Impacts of a cyclonic sub-mesoscale eddy on Synechococcus population dynamics and picoeukaryote distributions in the Western Mediterranean Sea.*
2. **Abbott K.**, Mahadevan A. (in prep) *Tracer distributions capture timescales of submesoscale dynamics.*
3. **Abbott K.**, Matantseva, O., Spezzano, R., Wittmers, F., Van Mooy, B., Worden, A.Z., Mahadevan, A. (in review at Nature Geoscience). *Microbial respiration controls the fate of subducted organic carbon in the mesopelagic ocean.*

Peer-reviewed journal publications

1. Middleton L., Wu, W., Johnston, T.M.S., (and 23 additional authors, including **Abbott K.**) (2025). *Ocean cyclone splitting ventilates the upper ocean*. Science Advances, 11(30). <https://doi.org/10.1126/sciadv.adu3221>
2. **Abbott K.**, Mahadevan, A. (2024). *Why is the monsoon coastal upwelling signal subdued in the Bay of Bengal?* Journal of Geophysical Research: Oceans, 129(12). <https://doi.org/10.1029/2024JC022023>
3. Grone, J., Poirier, C., **Abbott K.**, Wittmers, F., Jaeger, G.S., Mahadevan, A., Worden, A.Z. (2024). *A single Prochlorococcus ecotype dominates the tropical Bay of Bengal with ultradian growth*. Environmental Microbiology, 26(3). <https://doi.org/10.1111/1462-2920.16605>
4. **Abbott, K.**, Richards, M. (2020). *Elastic flexure of young, overlapping basaltic lava flows offshore the Galápagos and Hawaiian islands: Observations, modeling, and thermal/chronological analysis*. Geophysics, Geochemistry, Geosystems, 21(3). <https://doi.org/10.1029/2019GC008864>

Non-peer-reviewed publications

1. **Abbott K.**, “Setting sail during the Southwest Monsoon” Woods Hole Oceanographic Institution. <https://www.whoi.edu/setting-sail-during-the-southwest-monsoon/>
2. Middleton, L., **Abbott K.**, Berta, M., Mahadevan, A. (2025). “Data Report: Observations from the CALYPSO 2022 Field Campaign.” Woods Hole Oceanographic Institution. <https://doi.org/10.1575/1912/71916>
3. **Abbott K.**, Middleton, L., Wittmers, F., Grone, J., Matantseva, O., Carlson, C., Halewood, E., Achterberg, E., Riebesell, U., Mahadevan, A. (2025). “CALYPSO 2022 Biogeochemical Data Report.” Woods Hole Oceanographic Institution. <https://doi.org/10.1575/1912/71917>

SELECTED PRESENTATIONS (FIVE MOST RECENT)

1. **Abbott, K.** (2026) *Microbial respiration controls the fate of subducted organic carbon in the mesopelagic ocean*. Invited seminar at UMass Dartmouth’s School for Marine Science & Technology.
2. **Abbott, K.**, Mahadevan, A. (2025) *Why is the monsoon coastal upwelling signal subdued in the Bay of Bengal?* Invited seminar for EKAMSAT Science Seminar Series, virtual.
3. **Abbott, K.**, Matantseva, O., Spezzano, R., Wittmers, F., Van Mooy, B., Worden, A.Z., Mahadevan, A. (2024) *Biological Transformations During Submesoscale Subduction*. Talk at the Ocean Sciences Meeting; New Orleans, LA.
4. Mahadevan A., **Abbott, K.**, He, J. (2024) *Suppression of coastal upwelling*. Talk at the Ocean Sciences Meeting; New Orleans, LA.
5. **Abbott, K.**, Matantseva, O., Spezzano, R., Van Mooy, B., Worden, A., Mahadevan, A. (2023) *Observations of microbial community respiration in the Balearic Sea*. Poster at the Ocean Twilight Zone Symposium; Woods Hole, MA.
6. **Abbott, K.**, Matantseva, O., Spezzano, R., Worden, A., Mahadevan, A. (2023) *Phytoplankton community composition and response to submesoscale subduction in the Balearic Sea*. Poster at the Ocean Carbon & Biogeochemistry Summer Workshop; Woods Hole, MA.

TEACHING

WHOI high-performance computing workshop *Fall 2024*
(co-organized with Anthony Meza)

MIT 12.800 Fluid Dynamics of the Atmosphere and Ocean *Fall 2023*
Teaching assistant for Drs Amala Mahadevan and Claudia Cenedese
• Overall rating 7.0/7.0, 14 respondents

WHOI summer math review lecturer *Summer 2021, 2022, 2023*
Multivariable Calculus, Ordinary Differential Equations

The Carpentries workshop teaching assistant *2020 - 2024*

- Summer 2024 (2-day workshop): Taught incoming WHOI students and staff introductory programming in Bash and Python
- Summer 2021 (4-day workshop): Taught incoming WHOI undergraduate and graduate students introductory programming in Bash and Python
- Fall 2020 (1-day workshop): Taught WHOI technical and scientific staff how to use Python to analyze geospatial data

American Museum of Natural History instructor, BridgeUP STEM *2018 - 2019*

- Curriculum available at <https://github.com/amnh/BridgeUP-STEM-Oceans-Six>

FIELDWORK

TN-445 cruise (Bay of Bengal) *June 26 - Aug. 7, 2025*

- Meteorological data, aerobic respiration measurements and microbial community sampling

AR-80 Joint Program student cruise (New England shelf break) *Nov. 17 - 19, 2023*

- Student technical lead, biological sampling

CALYPSO ONR DRI cruise (Balearic Sea) *Feb. 17 - March 11, 2022*

- Biogeochemical sampling co-lead

NASA Sub-Mesoscale Ocean Dynamics Experiment pilot cruise (San Francisco coast) *Oct. 19 - Nov. 8, 2021*

- Biogeochemical sampling and underway CTD operations

LEADERSHIP AND SERVICE

- Reviewer for Geophysical Research Letters, JGR Oceans, Nature Communications, Journal of Marine Systems

Seminars and meetings

• **Co-organizer, MIT Program in Atmospheres, Oceans and Climate Colloquium** *2022-23*
• **Co-organizer, CALYPSO DRI fall meeting** *September 2022*

Diversity, Equity and Inclusion

• **Joint Program Applicant Support & Knowledgebase (JP ASK) board member** *2022 - 2025*

WORKSHOPS

Ocean Biogeochemical Virtual Institute workshop on primary production (Schmidt Sciences; Honolulu, HI)	<i>Dec. 2025</i>
Regional ocean modeling with MOM6 in the Community Earth System Model framework (NCAR; Boulder, CO)	<i>Oct. 2025</i>
How Viruses Affect Carbon Storage in the Ocean (Radcliffe Institute at Harvard University; Cambridge MA)	<i>March 2023</i>

OUTREACH

MIT Ocean Science and Engineering Systems Forum panelist	<i>2024</i>
Penikese Island School Science + Nature Summer Camp	<i>July 2022, '23, '24, '26</i>
Ocean HackWeek, virtual	<i>August 2020</i>
NYC Astronomy on Tap talk on detecting ocean carbon from space	<i>2019</i>
Watershed Stewards Program (AmeriCorps)	<i>2017-2018</i>
• Taught three classrooms (more than 60 students) a six-week curriculum about environmental science and conservation • Led outdoor education activities for local elementary and middle schools in Klamath-Trinity River and Humboldt communities	

MENTORING

Peer mentor for MIT-WHOI PhD student A. Scheffler	<i>2023 - present</i>
JP ASK student mentor (10)	<i>Fall 2020 - present</i>
Peer mentor for MIT EAPS undergraduate students (2)	<i>2021 - 2023</i>
Peer mentor for WHOI Summer Student Fellow (2)	<i>2022, 2024</i>

AWARDS AND HONORS

MIT EAPS Award for Excellence in Teaching	<i>2024</i>
NASA group achievement award, awarded to S-MODE science team	<i>2024</i>
AMNH Hackathon “Most innovative solution”	<i>2019</i>
Society of Professional Journalists finalist for news reporting for reporting on sexual harassment policy violations in UC system, with M. Wen and A. Yoon-Hendricks	<i>2017</i>
Phi Beta Kappa	<i>2017</i>
Summer of Applied Geophysical Experience	<i>2016</i>